SVAGA: Lab Test Results

SJSU lab, 1 x RTX 4090 (24 GB) | Prepared for team discussion

Why we ran these tests

The professor wants four fine-tuned local LLMs, at least 10,000 training examples, and results on trusted public benchmarks. Before choosing models, dataset size and training budget, we measured on the lab GPU: does each candidate fit, how fast does it train, and how long an example can it handle. Everything below was measured today, except where marked as an estimate.

Terms in plain words

- **Fine-tuning (LoRA / QLoRA):** we do not retrain the whole model. We shrink it to 4-bit so it fits in GPU memory and train a small add-on file (an *adapter*). The adapter is small, reusable across the whole project, and can be uploaded to Hugging Face. We must record which exact base model it belongs to.
- **Token:** a small piece of text, about 3/4 of a word. 4k, 8k, 16k tokens = 4,096 / 8,192 / 16,384 tokens, roughly 6 / 12 / 24 pages of text. One training example is a whole agent session (task, commands, outputs), so it can be long.
- **Sequence length cap:** the longest example we train on. Longer examples need more GPU memory and time. Anything above the cap is cut or dropped.
- **MoE vs dense:** a Mixture-of-Experts model (like Qwen3-Coder-30B) has 30B parameters in total but only uses about 3B for each token, so it trains fast. A dense model (like Qwen3.6-27B) uses all its parameters every time, so it is slower.
- **Tokens/s:** training speed. Higher is better; it decides how long training takes.

Measured results (LoRA, 4-bit, Unsloth)

Model	Type	Load memory	Train speed (tokens/s)	Longest length that worked	Status
Qwen3-Coder-30B-A3B	MoE, 30B total / ~3B active	16.8 GB	about 2,300 - 2,800	16k (peak up to ~22 GB at 24k; 32k ran out of memory)	Works
Qwen3.6-27B	Dense, 27B	16.1 GB	2k: 783 4k: 827 8k: 818 16k: 787	16k (peak 21.7 GB)	Works
gpt-oss-20b	MoE, 21B total / 3.6B active	12.6 GB	2k: 1,583 4k: 1,375 8k: 1,111	8k (peak 19.3 GB); 16k ran out of memory	Works up to 8k
Devstral Small 2 (24B)	Dense, 24B	n/a	n/a	n/a	Does not load: official weights are FP8, which Unsloth cannot train in 4-bit

Generation (the model writing answers) with plain Hugging Face is slow: about 10 tokens/s on Qwen3.6-27B and about 19 tokens/s on gpt-oss-20b. Benchmarking will need a proper inference server (such as vLLM), which we have not tested yet.

What this means for training time (estimate)

Assumption: 10,000 examples at about 10k tokens each is roughly 100 million tokens for one pass. The real number depends on how we filter and truncate, so treat this as a rough guide.

Model	Approx. time for 100M tokens, one pass
Qwen3-Coder-30B	about 11 hours
Qwen3.6-27B	about 35 hours
gpt-oss-20b	about 18 - 25 hours
Fourth model	not chosen yet (a dense model of this size would be about 30+ hours)

Total is about 3 to 4 days of GPU time for four models on one machine, about half that with two machines. A reservation allows up to 2 days in a row. Using shorter examples or fewer tokens makes this much cheaper.

Dataset observations

- OpenThoughts-Agent-SFT 10K: examples are long (median about 15k tokens) and include timeouts and sub-traces. We have to choose between dropping long rows and truncating them.
- A smaller cap (for example 8k so that gpt-oss fits) loses more of the data. If models get different caps, the comparison is not equal; this must be decided deliberately.
- Proposed mix (about 70% OpenThoughts, 20% WebGen-Instruct, 10% SafeCoder) is closer to prompt-to-app, but we have not yet checked what WebGen and SafeCoder rows actually contain (prompts only, or full solutions), or the SecureCode licence (non-commercial). Benchmark test splits must stay out of training.

Problems found

- **Docker is not available** inside the lab container (no docker command, no socket). Harbor and Terminal-Bench need Docker, so the benchmark cannot run there as is. Options: ask the lab admin about the Docker socket, use a cloud sandbox, or run only the benchmark on another machine that has Docker. Not verified yet.
- **Fast inference (vLLM)** is untested. It is needed for evaluation.
- **Devstral Small 2** does not load with our setup. A different route (plain transformers with FP8 to 4-bit conversion) is uncertain.

Decisions to take (nothing is decided yet)

- **Four models.** Measured so far: Qwen3-Coder-30B, Qwen3.6-27B, gpt-oss-20b. The fourth is open. Aim for at least two different model families, chosen from published scores (SWE-bench / Terminal-Bench), licence, and fit on 24 GB.
- **Dataset mix, sequence cap, number of examples per model.**
- **Where to run the benchmark** (Docker problem) and how to serve models for evaluation.
- **Questions for the TA / professor:** Terminal-Bench version and expected baselines; whether training on a subset counts toward the 10,000 examples.

Demo 2 reminders

Demo 2 is graded on ML modeling (3), ML evaluation (3), web system design and demo (4), and presentation (2.5), with slides and a demo video. The web app is the largest slice and does not need the lab GPU, so it can start now. Workbook 2 is due the day before Demo 2. Lab folders are wiped when a reservation ends: adapters and results must be copied out or uploaded to Hugging Face first.